Unit ⇒ 4

Field effect Transistor...

⇒ A field effect transistor is a three-terminal (drain, source and gate) semiconductor device in which current conduction is by only one type of majority carriers (electrons in case of n-channel FET and holes in case of P-channel FET). It is also called unipolar transistor.

Unlike a bipolar transistor FET requires small input current and gives an extremely high input impedance is the most important advantage over a BJT. Either BJT or FET devices can be used to operate in amplifier circuits.

TYPES OF FET

FET (Field effect transistor)

- JFET (Junction field effect transistor)
  - N-channel
  - P-channel
- MOSFET / IGFET
  - MOSFET (metal oxide semiconductor field effect transistor)
    - D-MOSFET (Depletion type MOSFET)
      - N-channel
      - P-channel
    - E-MOSFET (Enhacement type MOSFET)
      - N-channel
      - P-channel
  - IGFET (Insulated gate field effect transistor)

→ Difference between BJT and FET:-

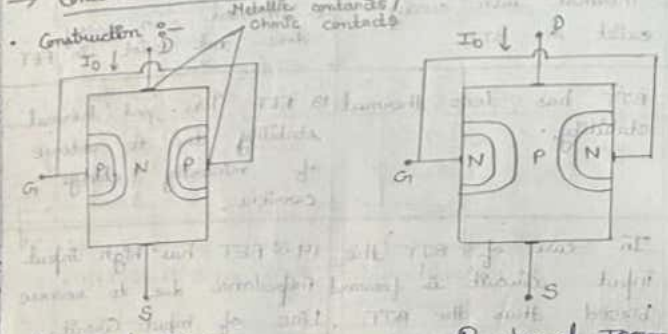
S.No.	BJT (Bipolar Junction Transistor)	FET (Field effect transistor)
1.	A type of transistor which uses two types of charge carriers i.e. electrones and holes for conduction.	A type of transistor in which electric field is used to control the flow of current in a semiconductor.

2.	In BJT the current flow is due to both majority and minority charge carriers thus it is a bipolar device.	2. In FET the electric current flow only due to majority charge carriers thus it is unipolar device.
3.	BJT has three terminals i.e emitter, base and collector.	3. FET also has three terminals i.e. source, drain and gate.
4.	BJT consist of two P-N junctions i.e. Emitter - base junction and Collector - base junction.	4. FET does not have P-N junctions.
5.	BJT is a current controlled device.	5. FET is a voltage controlled device.
6.	BJT are of two types :- NPN and PNP.	6. FET are also of two types :- N- channel and P- channel.
7.	BJT has three configuration:- Common - emitter, Common - base, Common - collector.	7. FET are also of three types :- Common - source, Common - gate and Common - drain.
8.	BJT is large in size enhance requires more space therefore it is more complicated to fabricate on IC (Integrate circuits).	8. FET is comparatively smaller in size hence it is easier to fabricate on IC.
9.	BJT is more sensitive to the changes in the applied voltage.	9. FET is less sensitive to the variations in the applied voltage.
10.	BJT has linear relationship b/w input and output.	10. FET has non- linear relationship b/w input and output.
11.	BJT has more thermal noise.	11. FET has much lower thermal noise.
12.	Thermal run away exist in BJT.	12. Thermal run away does not exist in FET.
13.	BJT has less thermal stability.	13. FET has good thermal stability due to absence of minority charge carriers.
14.	In case of BJT the input circuit is forward biased thus the BJT has low input impedance.	14. FET has high input impedance due to reverse bias of input circuit.
15.	BJT is suitable for low current applications.	15. FET is suitable for high current applications.
16.	The switching speed of BJT is low.	16 The switching speed of FET is high.

17.	BJT is cheaper to manufacture.	FET is expensive to manufacture.
18.	BJT is used as switch (Insaturation and cutoff mode) and amplifier (in active mode).	18. FET is used as switch (Inohmic and cutoff) and as amplifier (Insaturation mode).

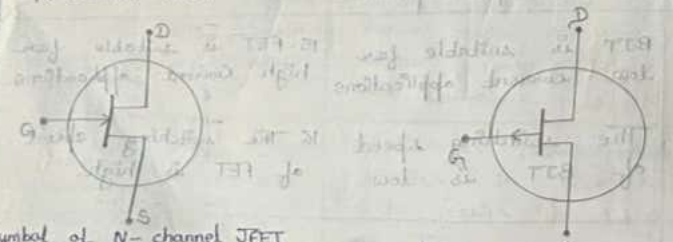
→ Construction and working of JFET:-

- Construction :-



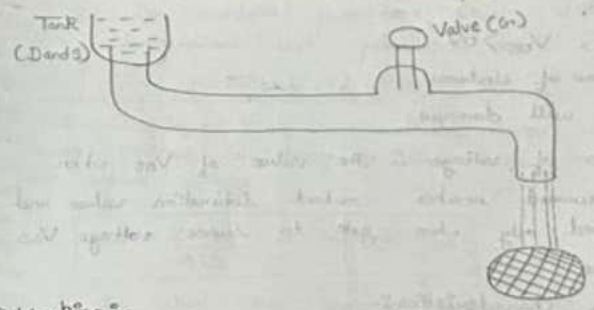
N-channel JFET

P-channel JFET



Symbol of N-channel JFET

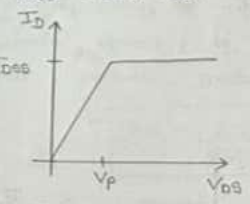
Symbol of P-channel JFET



→ Working :-

⇒ Case 1:-

$V_{GS} = 0V$, $V_{DS} > 0V$

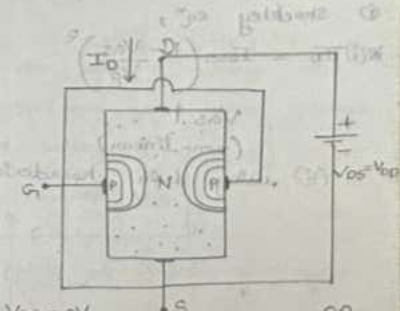


Output characteristics

or

Drain characteristics

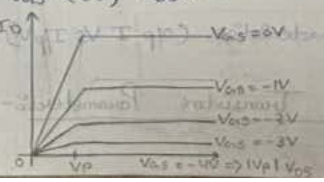
* V_P = pinch of voltage

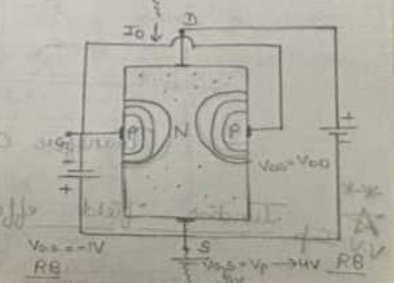


RB

⇒ Case 2:-

$V_{GS} < 0V$, $V_{DS} > 0V$





RB

RB

$V_{GS} > 0V$, $V_{DS} > 0V$

Heavy flow of electrons

Transistor will damage.

=> The pinch of voltage is the value of V_{DS} when drain current reaches constant saturation values and is defined only when gate to source voltage V_{GS} is zero.

* Transfer Characteristics :-

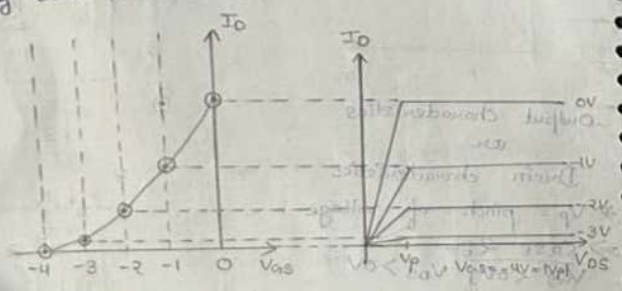
① Shockley eqⁿ,

*(i) $I_D = I_{DSS}\left(1 - \frac{V_{GS}}{V_P}\right)^2$

$V_{GS} \uparrow$

(non-linear)

(ii) using drain characteristics



Transfer Characteristics (O/P I Vs I/P V)

**

☆ V.V. Imp Junction field effect Transistor Parameters :-

=> JFET has certain parameters which determine the performance such parameters are :-

(I) A.C Drain Resistance (r_d) :-

=> It is defined as the ratio of change in drain-source voltage to change in drain-current at constant gate-source voltage.

$$r_d = \frac{\Delta V_{DS}}{\Delta I_D} \text{ at Constant } V_{GS}$$

It is also called the dynamic drain resistance.

(II) Transconductance (g_m) :-

=> It may be defined as the ratio of change in drain current to the change in gate-source voltage at constant Drain-source voltage.

$$g_m = \frac{\Delta I_D}{\Delta V_{GS}} \text{ at Constant } V_{DS}$$

It is measured in mA / Volt or micro-siemens.

(III) Amplification factor (μ) :-

=> It is defined as the ratio of change in drain-source voltage to the change in gate-source voltage at constant drain current.

$$\mu = \frac{\Delta V_{DS}}{\Delta V_{GS}} \text{ at Constant } I_D$$

$$\mu = \frac{\Delta V_{DS}}{\Delta I_D} \times \frac{\Delta I_D}{\Delta V_{GS}}$$

$$\boxed{\mu = r_d \times g_m}$$

(iv) D.C drain Resistance (R_{DS}) :-

=> It is also called the static or ohmic resistance of the channel and it is defined as the ratio of drain-source voltage to drain-current.

$$\boxed{R_{DS} = \frac{V_{DS}}{I_D}}$$

Que:- When drain-source voltage is changed by 1.5 Volts. The change in drain current is of 120 micro ampere. The gate-source voltage remaining unchanged determine the A.C drain resistance of the JFET.

=> $\Delta V_{DS} = 1.5\ V$

$\Delta I_D = 120\ mA = 120 \times 10^{-6}$

$r_d = ?$

$$r_d = \frac{\Delta V_{DS}}{\Delta I_D}$$

$$r_d = \frac{1.5}{120 \times 10^{-6}}$$

$$\boxed{r_d = 12.5 \times 10^{3}}$$

→ MOSFET (Metal oxide Semiconductor field effect transistor) :-

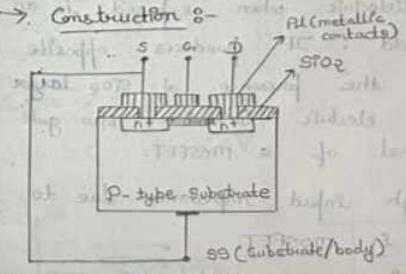
OR

IGFET (Insulated gate field effect transistor)

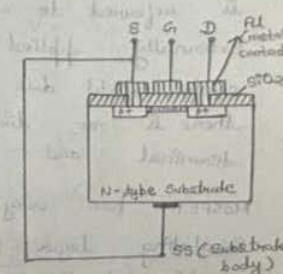
- → Depletion type MOSFET (D-MOSFET) / Depletion enhancement type MOSFET (DE-MOSFET)
- → Enhancement type MOSFET (E-MOSFET).

1. Depletion type MOSFET (D-MOSFET) / Depletion enhancement type MOSFET (DE-MOSFET).

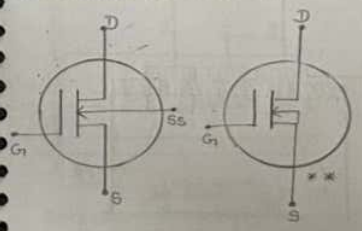
→ Construction :-

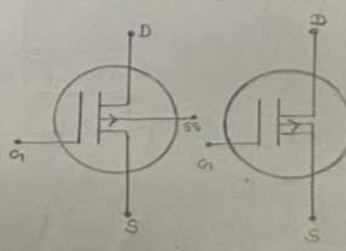


N-channel D-MOSFET



P-channel D-MOSFET

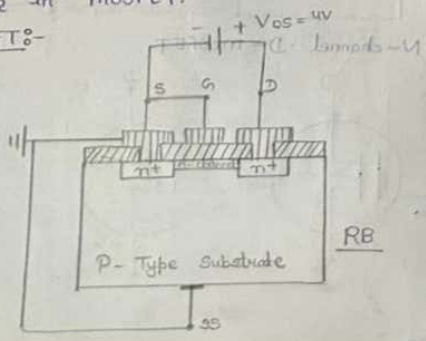




⇒ Para Construction of n-channel MOSFET. Generally take substrate of p-type. Substrate is the base or foundation on which device is constructed. Substrate is internally connected to source terminal the drain and the source terminals are connected through metallic contacts to n^+ dopped regions are liked by an n-channel the gate is also connected to metallic contacts but it remains insulated from n-channel by presence of thin SiO_2. It is a particular type of insulator, it is referred to a dielectric when exposed to an externally applied field. It produces opposite electric field due to the presence of SiO_2 layer there is no direct electric connection b/w gate terminal and channel of a MOSFET.

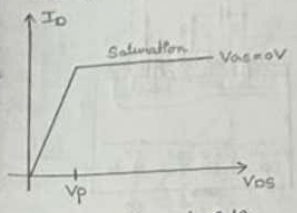
MOSFET has very high input impedance due to insulating layer SiO_2 in MOSFET.

→ Working of D-MOSFET:-

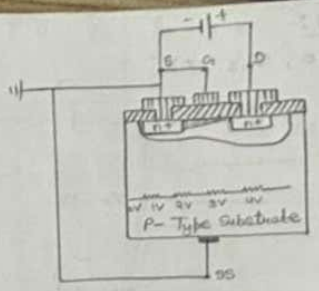


Case 1:-

$V_{DS} > 0V$, $V_{GS} = 0V$



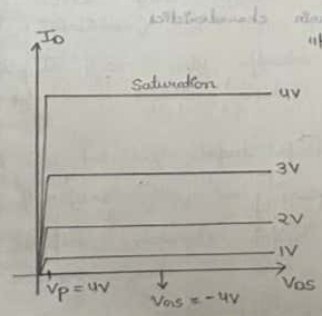
Output characteristics
or
drain characteristics



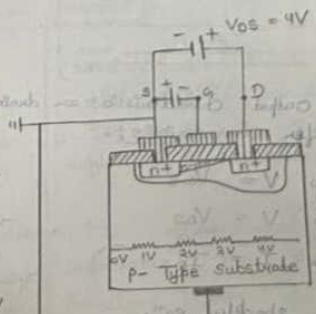
o/p I = ID
o/p V = VDS
I/p V = VGS

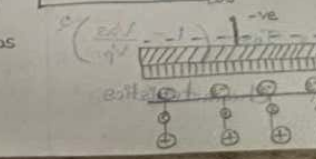
Case 2:-

$V_{DS} > 0$, $V_{GS} < 0V$



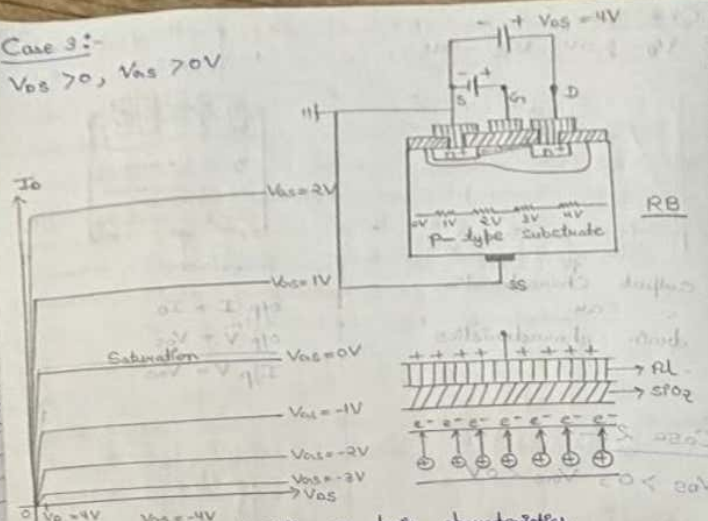
Output characteristics
or
drain characteristics
$V_{GS} = V_p$





Case 3:-

$V_{DS} > 0$, $V_{GS} > 0V$



Output Characteristics or drain characteristics.

→ Transfer Characteristics :-

I/P $V = V_{GS}$

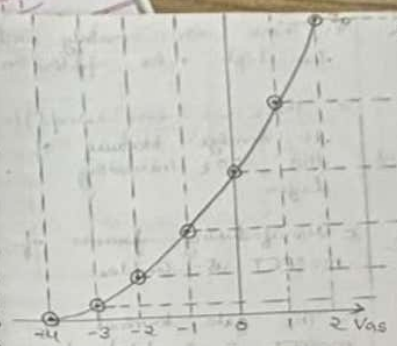
O/P $V = V_{DS}$

O/P $I = I_D$

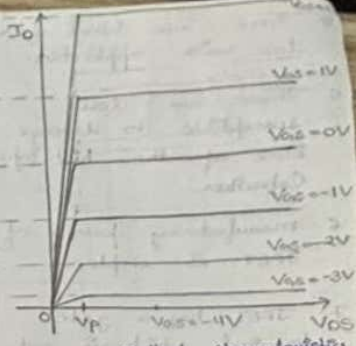
① By shockley eqn,

$$I_D = I_{DSS}\left(1 - \frac{V_{GS}}{V_P}\right)^2$$

② Drain Characteristics



Transfer Characteristics



Output characteristics or drain characteristics

→ Difference b/w JFET and MOSFET :-

JFET (Junction field effect transistor)	MOSFET (metal oxide semiconductor field effect transistor)
1. It can only operates in the depletion mode.	1. It operates in both depletion and enhancement mode
2. It has high input impedance therefore they are more sensitive towards input voltage signals.	2. It offers even higher impedance then the JFET therefore they are more sensitive.
3. It is relatively cheaper than MOSFET.	3. It is expensive than JFET.

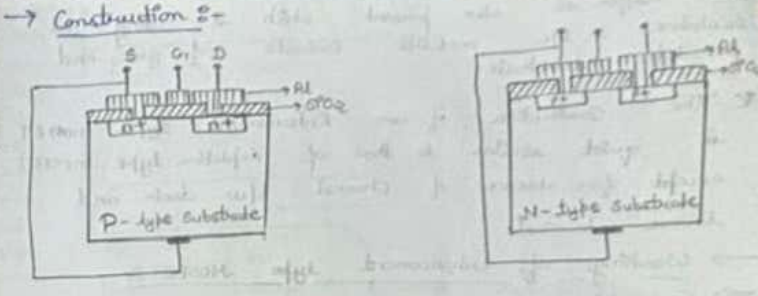
4. These are ideal for low noise application.	4. These are mainly used for high noise application.
5. These are less susceptible to damage because of the high input Capacitor	5. These are more susceptible to damage because of the SiO_2 insulating layer.
6. Manufacturing process of JFET is simple.	6. Manufacturing process of MOSFET is complex.
7. JFET has a gate terminal which is not insulated from the channel.	7. The gate terminal of MOSFET is insulated from thin layer of SiO_2
8. For JFET Gate Current is more.	8. For MOSFET Gate Current is less.
9. JFET has a Continuous channel. The channel exists permanently.	9. MOSFET has a Continuous channel only in depletion type but not in enhancement type.
10. In JFET the conductivity of the device is Controlled by the Reverse biasing of the gate.	10. In MOSFET the conductivity is Controlled by the charge carriers induced in the channel.

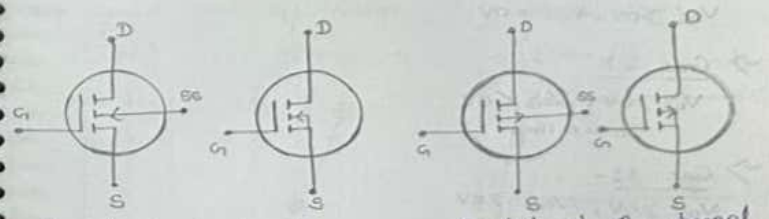
→ Enhancement type MOSFET :-

OR

E- MOSFET :-

→ Construction :-





Symbol of N- channel E- MOSFET

Symbol of P- channel E- MOSFET

⇒ A slab of p- type material is formed from a silicon base. It is referred to as substrate. Substrate is the supporting material upon or within which on IC is fabricated. The source and drain terminals are conneded through metallic Contacts (Al) to n^+ dopped regions. This is the primary difference b/w Construction of D- type and E- type MOSFET.

SiO_2 layer is also present which is acting as insulator in b/w metallic contacts of gate and p- type substrate.

* The construction of an Enhancement type MOSFET is quite similar to that of depletion type MOSFET except for absence of channel b/w drain and source terminal.

→ Working of enhancement type MOSFET:-

⇒ Case 1:-

$V_{DS} > 0V$, $V_{GS} = 0V$

⇒ Case 2:-

$V_{GS} > 0V$, $V_{GS} < 0V$

$I_D = 0$ Amp.

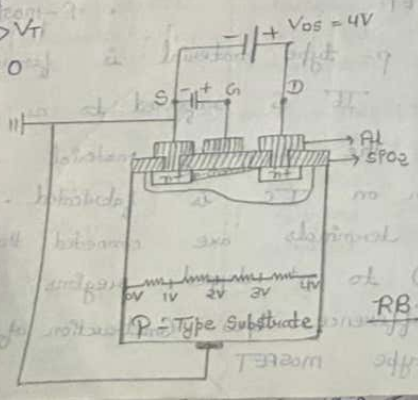
⇒ Case 3:-

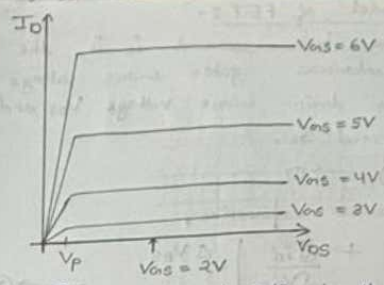
$V_{DS} > 0V$, $V_{GS} > 0V$

Let $V_T = 2V$

$V_{GS} > V_T$

$I_D \neq 0$





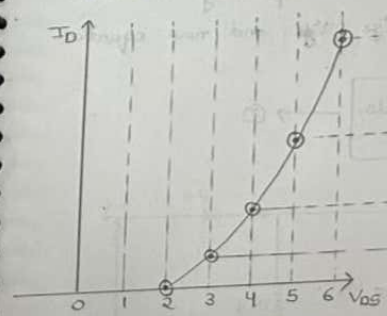
$2V = V_T$ (Threshold Voltag

$V_{GS} > V_T$

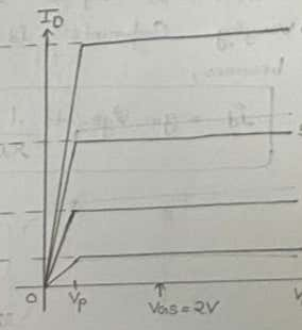
Drain characteristics / Output Characteristics

O/p Current and O/p Voltage, I/p Voltage Varies

(I_{DS}) (V_{DS}) (V_{GS})



Transfer characteristics



→ Two ways to draw transfer characteristics:-

① By shockley eqⁿ,

$$I_D = I_{DSS}\left(1 - \frac{V_{GS}}{V_P}\right)^2$$

② Drain Characteristics.

★ Small signal model of FET :-

⇒ In FET instantaneous drain current i_D is the function of the instantaneous gate-source voltage V_{GS} and instantaneous drain-source voltage V_{DS} and is therefore expressed as,

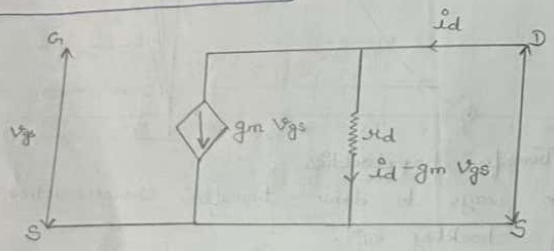
$$i_D = f(V_{GS}, V_{DS}) \longrightarrow ①$$

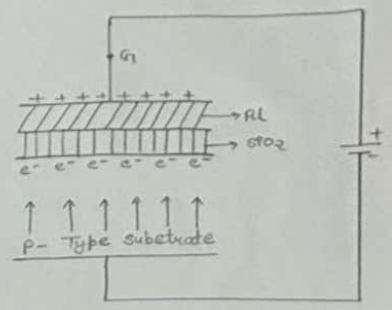
Applying Taylor's series expression,

$$\Delta i_D = \frac{\partial i_D}{\partial V_{GS}}\Big|_{V_{DS} \to \text{Constant}} \Delta V_{GS} + \frac{\partial i_D}{\partial V_{DS}}\Big|_{V_{GS} \to \text{Constant}} \Delta V_{DS} \longrightarrow ②$$

Using the conventional small signal notations Δi_D, ΔV_{GS}, ΔV_{DS} may be replaced respectively be time varying components i_d, V_{gs}, V_{ds} and now equation becomes,

$$\boxed{i_d = g_m V_{gs} + \frac{1}{r_d} V_{ds}} \longrightarrow ③$$



Low frequency small signal leaded of FET.

→ MOSCAP :-
(MOSFET Capacitor)